

Spring Free Build - Episode 1

Hello! You've been hired into W+ Software Solutions. We've got a couple different contracts that have to get done **right now**. Take a look at the customer requests below, pick one, and then build it. That's really it. No clarifications are available- just what the customer provided. You got this.

Colin County Public Library

Contract number: 00001

"Hi! We're a small public library and we're struggling to keep track of our books. Right now everything is on paper and it's a mess. We have two kinds of books — regular fiction novels and reference books like encyclopedias — and they each have slightly different information we want to record. We'd love a program where staff can add new books, remove ones we no longer carry, look up books by a specific author, and see a full list of everything we have. Nothing fancy, just something that works!"

What they need

- Both fiction and reference books share some common details — title, author, and a unique ID number. But each type also has its own extra info (fiction books have a genre; reference books have an edition number).
- Each book should be able to give a short description of itself — something a librarian could read at a glance to know what it is.
- Staff need to add a book to the collection, remove a book using its ID number, and see a list of everything currently in stock.
- Staff should be able to search for all books written by a particular author and see the results.
- When you're done, demo the system by adding a few books of both types, running a search, removing one, and printing the final collection.

Alicia and Tai's House of Peculiarities

Contract number: 00002

"Greetings. We operate a small but distinguished museum dedicated to haunted and cursed artifacts. We currently have two categories of exhibits: possessed objects — things like mirrors, dolls, and music boxes that contain a spirit — and cursed relics, which are ancient items that carry a hex or misfortune. Each type has its own particular details we track, but they all share basic information like a name, an origin date, and a catalog number. We desperately need a system to log new acquisitions, remove items that have been... returned to the void, look up artifacts by their origin era, and get a full inventory at any time. Our current system is a leather-bound book and frankly it keeps opening on its own."

What they need

- All artifacts share a name, an origin date, and a unique catalog number. Possessed objects also have the name of the spirit inside them, while cursed relics have a description of the misfortune they bring.
- Each artifact should be able to generate a dramatic exhibit label — the kind of thing you'd read on a placard next to a glass case.
- Staff need to log a new artifact, remove one by its catalog number, and view the full current inventory.
- Curators should be able to search the collection by era — for example, find all artifacts from the 1800s.
- Demo the system by adding a handful of artifacts of both types, searching by era, deaccessioning one, and printing the remaining collection.